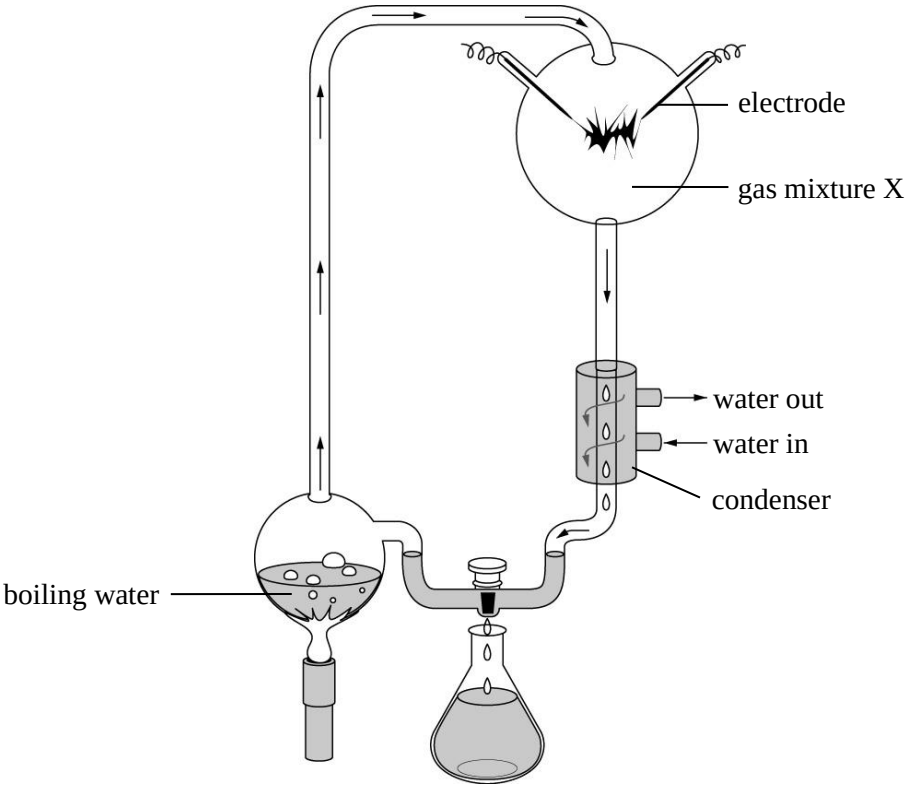
Short questions

|!|BCESC0300000001|!| **SI**

The diagram below shows the set-up of the Miller-Urey experiment in 1953.



(a) What condition of the early Earth is simulated by the gas mixture X? (1 mark)

(b) State ***two*** components of the gas mixture X. (2 marks)

(c) What does the result of the experiment above show about the origin of life on the early Earth? (3 marks)

##

(a) the primitive atmosphere (1)

(b) Any ***two***: (1, 1)

methane / ammonia / hydrogen / water vapour

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(c) Heating and lightnings provided energy to (1)
drive chemical reactions between the inorganic substances in the atmosphere, (1)
which produced simple organic molecules. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Understanding basic concepts	2
(c)	Applying concepts	3

|!|BCESB0300000002|!| ★ **Cross topic**

The photograph below shows a fossilized cone of an extinct species of conifers.



Source: https://en.wikipedia.org/wiki/File:Petrified_Araucaria_cone.jpg
Credit: Brocken Inaglory

(a) What is the function of cones in conifers? (1 mark)

(b) Fossils are commonly formed from the hard parts of organisms. Give ***two*** examples of such hard parts in animals. (2 marks)

(c) Why are the soft parts of organisms seldom fossilized? (2 marks)

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(d) Suggest ***one*** way by which the soft parts of organisms can be preserved. (2 marks)

##

(a) Cones enclose the naked seeds of conifers. (1)

(b) Any ***two***: (1, 1)
bones / teeth / shells / other reasonable answers

(c) The soft parts are easily broken down (1)
by decomposers (e.g. fungi and bacteria). (1)

(d) Being completely trapped in ice or resin (1)
At very low temperature / Without oxygen, the rate of decomposition is greatly slowed down. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Applying concepts	2
(d)	Applying concepts	2

|!|BCESB0300000003|!| ★

When an organism dies, its body begins to decompose. However under certain conditions, the dead remains can be preserved.

(a) Explain why the conditions in the following environments is suitable for fossil formation.

(i) Riverbed (3 marks)

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(ii) Polar regions (2 marks)

(b) Suggest ***two*** ways by which the fossil record is useful for studying the evolutionary relationship of organisms. (2 marks)

##

(a) (i) Dead remains may be quickly buried by mud or other sediments, and may not be eaten by other organisms. (1)
Sediments pile up and exert pressure on the skeleton. (1)
Minerals from the environment may fill the skeleton which slowly becomes fossilized. (1)

(ii) When dead remains are completely trapped in ice, the very low temperature (1)
greatly slows down the rate of decomposition. (1)

(b) Any ***two*** of the following: (2)

- The anatomy of different organisms can be compared by studying fossils.
- The morphology of different organisms can be compared by studying fossils.
- The time of existence of organisms / chronological order of existence of organisms can be deduced from the fossil record.

(Or other reasonable answers)

(7 marks)

##

	Skills assessed	Marks
(a)(i)	Applying concepts	3
(a)(ii)	Applying concepts	2
(b)	Understanding basic concepts	2

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|!|BCESC0300000004|!| **Cross topic**

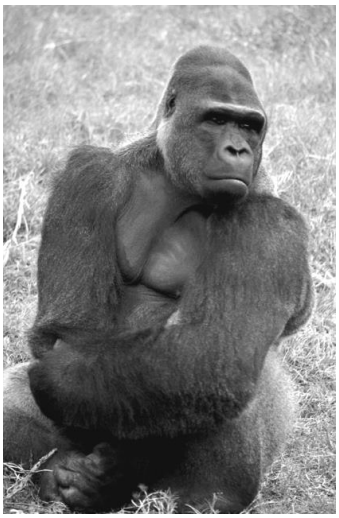
The following photographs show three types of animals: carp, dolphin and gorilla.



carp



dolphin



gorilla

(a) Identify the class to which each type of the animals belong. (2 marks)

(b) State ***one*** common characteristic shared by dolphins and carp that enables them to live in an aquatic environment. (1 mark)

(c) Although dolphins and carp live in the same habitat, the evolutionary relationship between dolphins and carp is more distant than the evolutionary relationship between dolphins and gorillas. Suggest ***one*** way to prove this. (1 mark)

##

(a) carp: fish (1)
dolphin and gorilla: mammals (1)

(b) Both dolphins and carp have fins for swimming / other reasonable answer (1)

(c) Any ***one*** of the following: (1)

- Comparing the DNA base sequences of the three organisms
- Comparing the amino acid sequences of common proteins in the three organisms

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(Or other reasonable answers)

(4 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	1
(c)	Understanding basic concepts	1

|!|BCESC0300000005|!|

Complete the following paragraph with suitable words. (5 marks)

There are different sources of evidence for evolution. If the remains or traces of organisms that lived in the past were preserved by natural processes, (a)_____ were formed.

(b)_____ is the study of the similarities and differences in physical structures of different organisms. Organisms with more similarities in their anatomy are considered to be more closely related. Structures from different species having a similar basic form but with different functions are called (c)_____. The possession of these structures by different species suggests that the different species have evolved from a (d)_____.

Some scientists believe that organisms with closer evolutionary relationships would have more genetic similarities and fewer differences. By comparing the base sequences of DNA, or the (e)_____ of common proteins, scientists can study the evolutionary relationship of different organisms.

##

(a) fossils (1)
(b) Comparative anatomy (1)
(c) homologous structures (1)
(d) common ancestor (1)
(e) amino acid sequences (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1

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(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

|!|BCESB0300000006|!| ★

The diagram below shows the bones that comprise the forelimb of a dolphin.



The photographs below show the wings of a bee and a bat respectively.



Source: https://commons.wikimedia.org/wiki/File:Bee_wings_in_detail.jpg



Source: https://commons.wikimedia.org/wiki/File:Bat-wing_underside.jpg

Explain whether the bee or the bat has a closer evolutionary relationship with the dolphin based on the structures of the dolphin's forelimb, the bee's wings and the bat's wings. (5 marks)

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##

The dolphin has a closer evolutionary relationship with the bat than the bee. (1)
Both the dolphin's forelimb and the bat's wings are supported by bones, (1)
but the bee's wings contain no bones. (1)
The dolphin's forelimb and the bat's wings are composed of similar bones constructed in similar ways. (1)
These similarities suggest that the dolphin and the bat may have evolved from a common ancestor. (1)

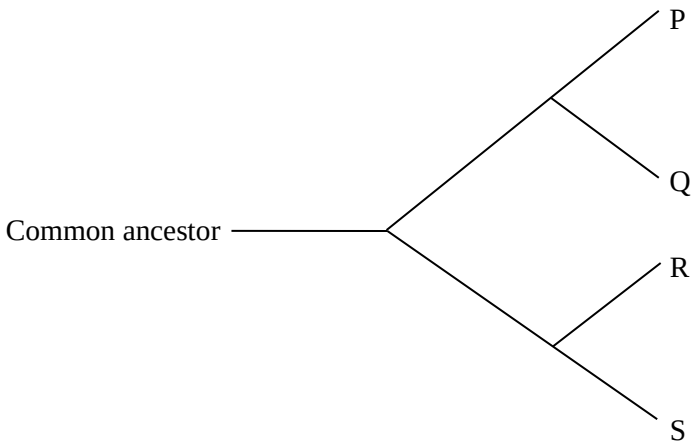
(5 marks)

##

Skills assessed	Marks
Analysing data	3
Applying concepts	2

|!|BCESB0300000007|!| ★

The following evolutionary tree shows the evolutionary relationship among four species (P, Q, R and S) of marsupials.



(a) Marsupials belong to a class of vertebrates. Describe ***two*** typical characteristics of this class of vertebrates. (2 marks)

(b) Which species on the evolutionary tree has the closest evolutionary relationship with species Q? (1 mark)

(c) State and explain how scientists can make use of molecular biology to study the evolutionary relationship among organisms. (3 marks)

##

(a) Any ***two*** of the following: (2)

- Mammals are endotherms.
- Mammals have hair covering their bodies.
- Female mammals have mammary glands, which produce milk after the birth of offspring.
- The skin of many mammals has sweat glands.

(Or other reasonable answers)

(b) Species P (1)

(c) Comparing the DNA base sequences of different organisms (1)

Comparing the amino acid sequences of common proteins in different organisms (1)

Organisms with closer evolutionary relationships would have more similarities and fewer differences in these sequences. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Analysing data	1
(c)	Understanding basic concepts	3

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|!|BCESA0300000008|!| ★★

The table below shows part of the base sequence of a gene common in three animals: the red panda, the Giant panda and the raccoon.

Animal	Base sequence of the gene
Red panda (*Ailurus fulgens*)	G A G T A C C A T T A C G C A C
Giant panda (*Ailuropoda melanoleuca*)	T A G T T C C A T T A C G C T G
Raccoon (*Procyon lotor*)	G A G T A C C A T T A C G C A T

(a) According to the table, which two animals are more closely related? (1 mark)

(b) Explain your answer in (a). (2 marks)

(c) Sketch an evolutionary tree of the three animals in the space below. (2 marks)

##

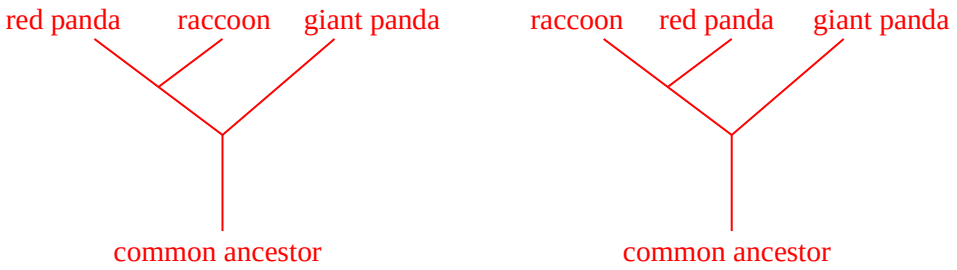
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(a) The red panda and the raccoon (1)

(b) There is only one base difference in the sequence of the gene between the red panda and the raccoon. (1)

The base sequence of the gene in the giant panda differs from those of the red panda and the raccoon by four bases respectively. (1)

(c)



or

Correct tree (1)

Correct labels (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	2
(c)	Presenting information	2

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